Is the transfer market all about big cheques?

€222M

Neymar, FC Barcelona → Paris Saint-Germain, 2017
the largest fee in the dataset (3,469,314 rows)

98.3%: zero fee

1.7%: fee > 0 (58,755 rows)

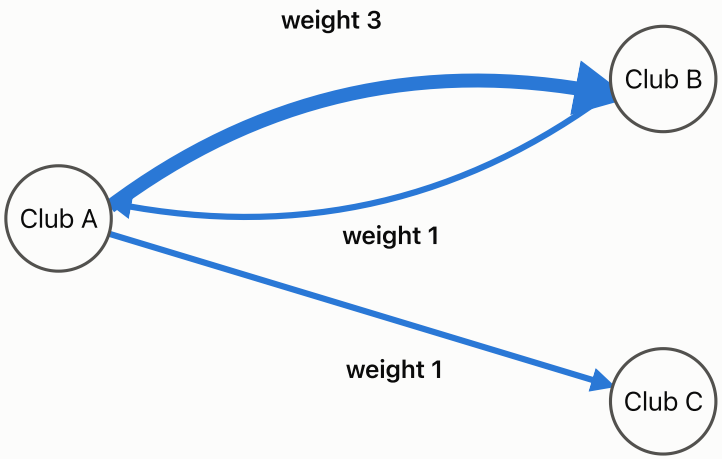
€0 = loan, end of loan, retirement, free transfer… or undisclosed fee

> **Which clubs occupy the most central positions in the observed network, and which groups of clubs mainly exchange players with each other?**

Source: results/data_inspection.md

Alessandro Arensberg · Charles Defosseux · Louis Allio — github.com/aarensberg/football-transfer-network-analysis

What does a link between two clubs mean?



Illustrative diagram: fictitious clubs.

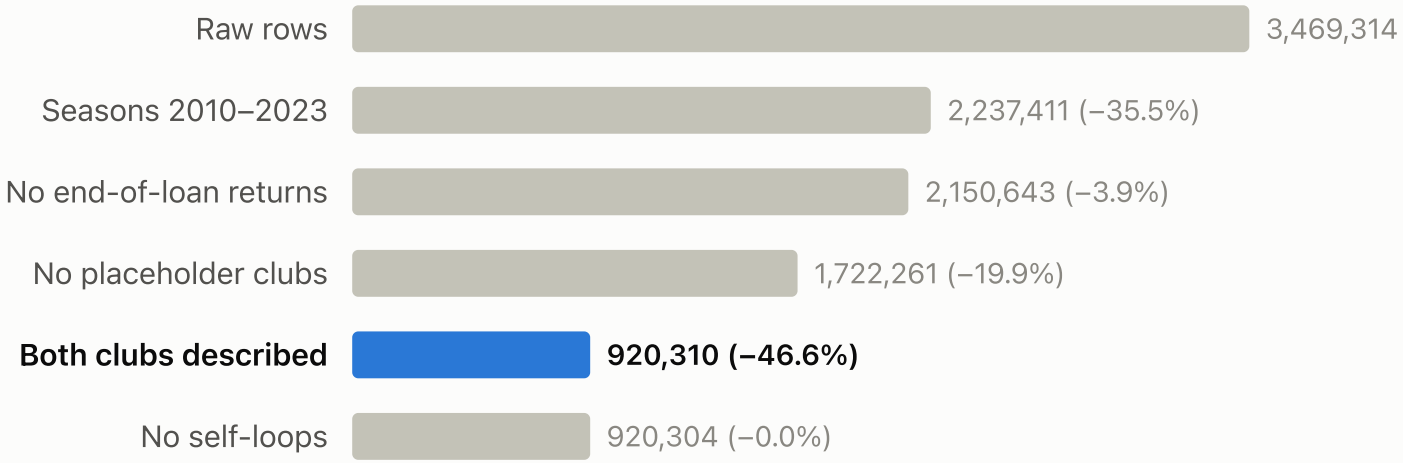
Node = a club

Edge = from the club a player leaves to the club he joins
Weight = number of transfers (not the fee, which is almost always zero)

directed · simple (one edge per pair of clubs) · weighted · no self-loops

Source: Kaggle, Otavio Brito (data scraped from Transfermarkt)

What is left after cleaning?



12,009 clubs **450,217** edges **920,304** transfers

the network observed in Transfermarkt (2010–2023), not the whole world of football

Source: results/graph_summary.md

Is the network all in one piece?

No: **6** weakly connected components, **404** strongly connected ones.

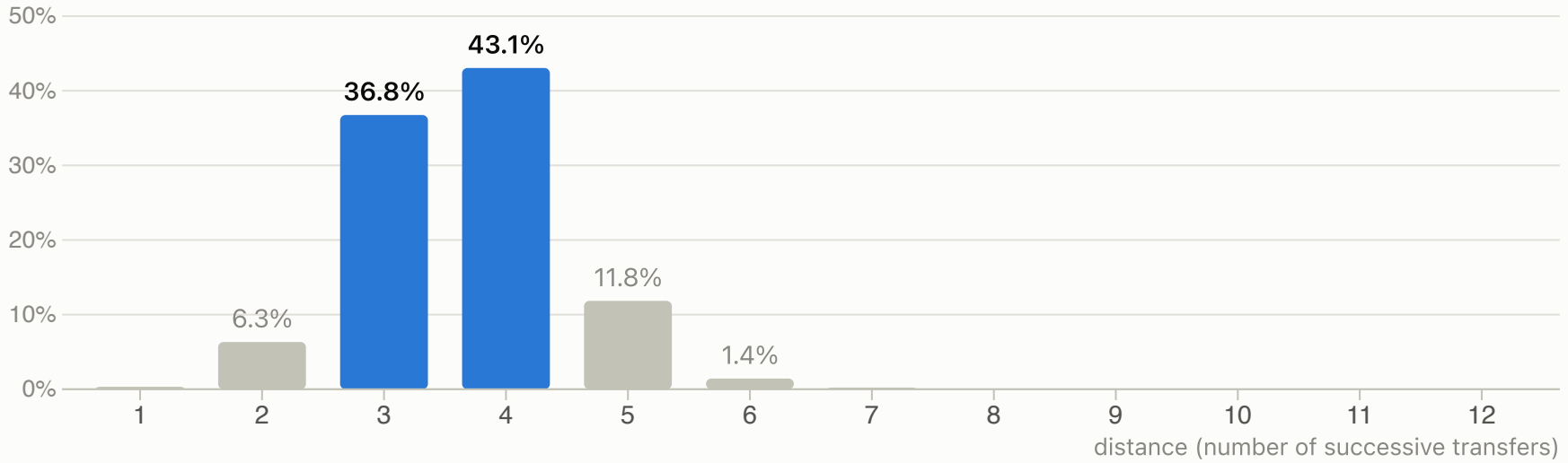


411 other clubs, in 403 small SCCs: 3.4%

The largest strongly connected component (SCC) carries **99.65%** of transfers → directed distances are computed on it.

Source: results/connectivity.md

How many transfers apart are two clubs?



From 9 hops on: at most 0.01% of pairs per distance, up to the diameter of 12.

Directed, giant SCC (11,598 clubs)
diameter **12** · radius **6** · mean 3.65 — in hops, exact computation

Undirected, vs random graph with the same degrees
mean distance 3.28 vs 2.74 · clustering **0.336** vs 0.015

Source: results/distances.md, results/structure.md

Which club is the most central?

About ten measures computed: **no club ranks first on all of them**.

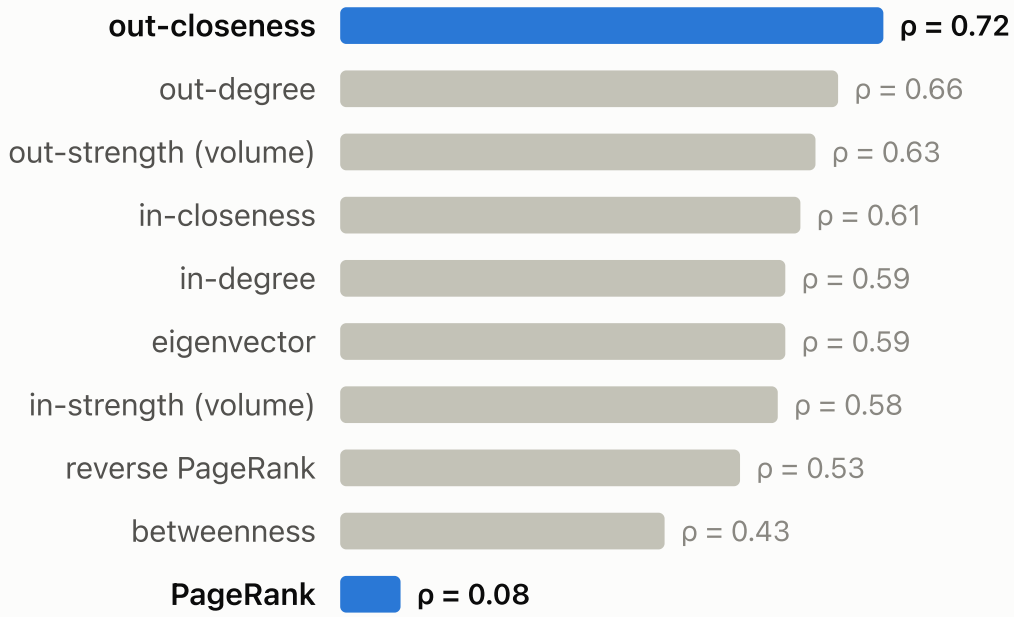
	out-strength (volume) outgoing transfers	out-closeness mean proximity to other clubs	PageRank where flows end up	betweenness presence on shortest paths
Parma	**1st**	11th	707th	14th
Hajduk Split	43rd	**1st**	5,487th	379th
Kalev Tallinn	2,593rd	1,060th	**1st**	13th
BSV Rehden	1,134th	579th	402nd	**1st**

Rank among 12,009 clubs; closeness: 11,598 clubs of the giant SCC. 1st = most central.

Source: results/tables/centrality.csv

Does PageRank point to the big clubs?

Rank correlation (Spearman) between each measure and the clubs' mean 2010–2023 market value (valued clubs only)

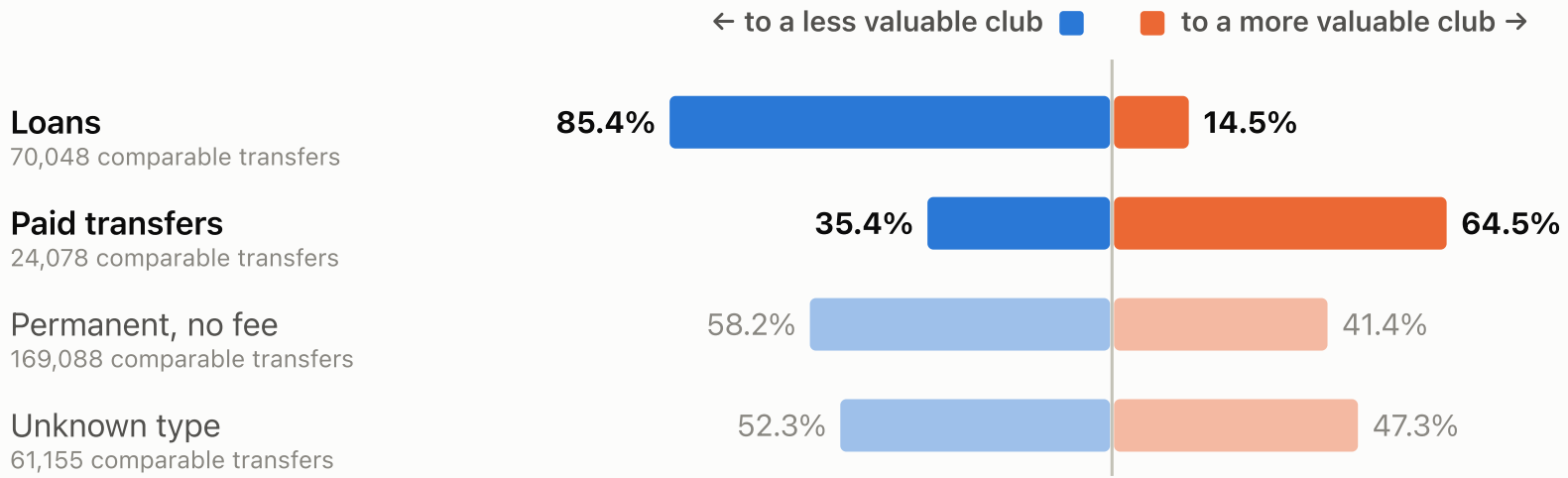


Correlation ≠ causation: being well connected does not make a club rich.

Source: results/centrality.md (external validation)

Which way do players move?

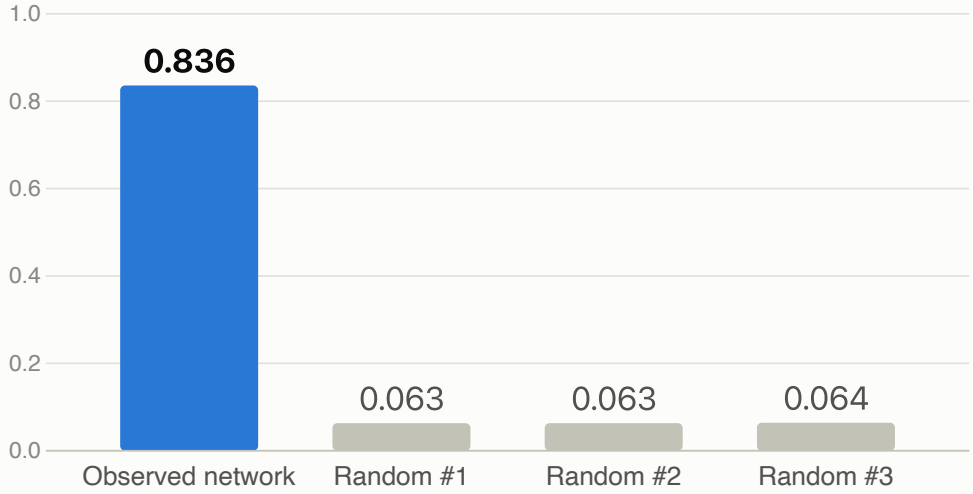
Market value of the buying club compared with the selling club, in the season of the transfer



Comparable: 35.2% of transfers (value known and > 0 for both clubs). We observe a direction, not an intention.

Source: results/hierarchy.md

Do these groups of clubs really exist?



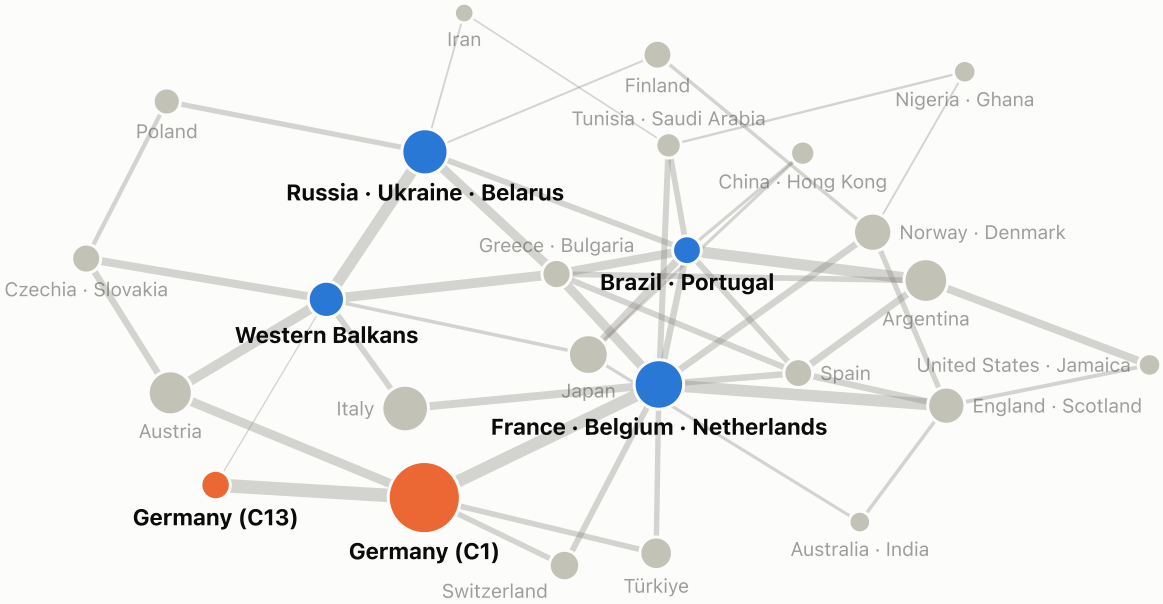
Louvain: undirected graph (weight = sum of both directions), resolution 1
→ **35 communities**, modularity **0.836**

Random networks that keep each club's activity (weighted configuration model) → **0.06**

Leiden: almost identical partition (34 communities, NMI 0.992)

Source: results/communities.md

Who trades with whom?



84.9% of transfers stay within one country

Louvain level 1: **389** groups, national purity 0.98 → **35** blocs, purity 0.68

Blocs mentioned: **France · Belgium · Netherlands**, **Brazil · Portugal**, **Russia · Ukraine · Belarus**, **Western Balkans** *shared language and history: our reading, not tested*

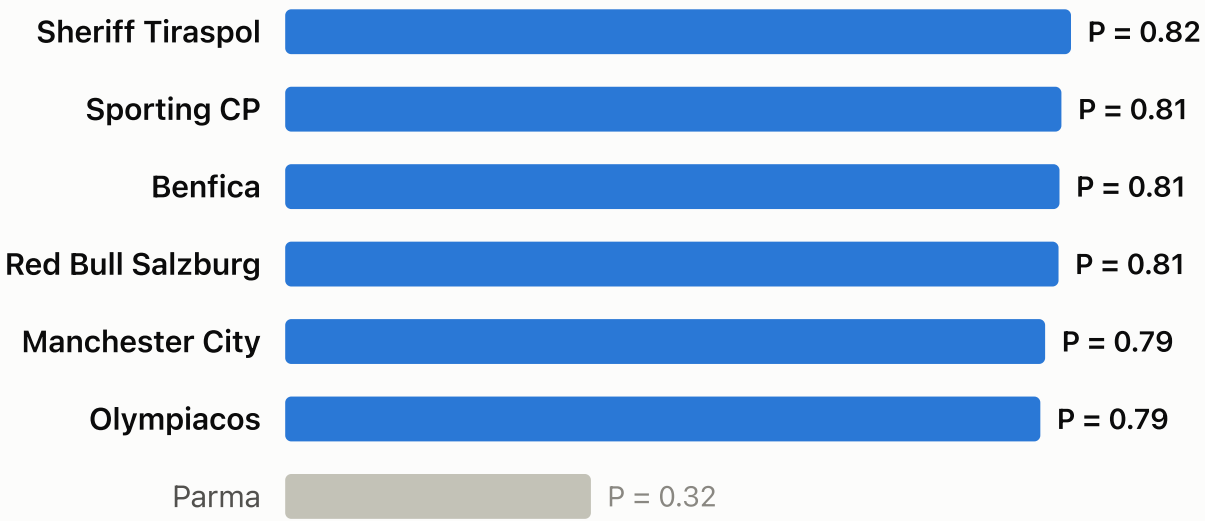
Germany split into two communities: probably an effect of the coverage of amateur leagues

25 largest communities; size ∝ number of clubs, width ∝ transfers; position = proximity in the network, not geography.

Source: results/tables/communities.csv, results/countries.md, results/communities.md

Who connects the blocs?

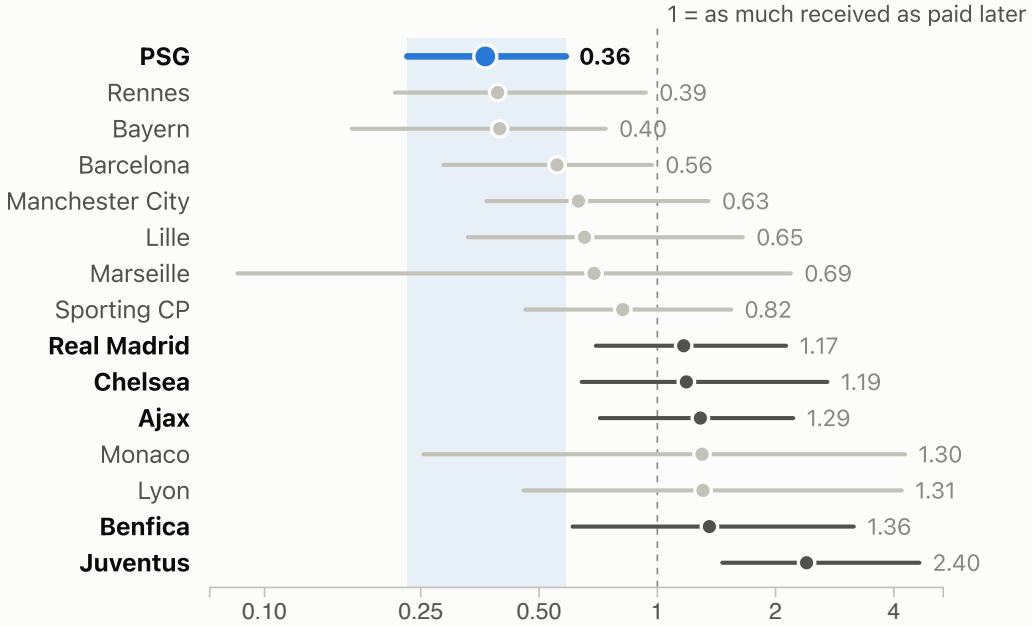
Participation coefficient P: close to 1 if a club's exchanges are spread over many communities, 0 if they stay within one (clubs with strength ≥ 100)



75.5% of the 1,000 edges with the highest betweenness link two communities, against 19.7% of all edges.

Source: results/tables/roles.csv, results/roles.md

Does PSG capture the value of the players it trains?



PSG, 215 departures evaluated: **€118.7M** received at departure / **€325.3M** paid later = **0.36**
Nkunku: €19.5M (Leipzig) → €60M (Chelsea)

Lowest of the 15 (point estimate); CI overlaps 9 others; 5 clearly above (in bold)

€0 = free or undisclosed: value realized elsewhere, not a "loss"

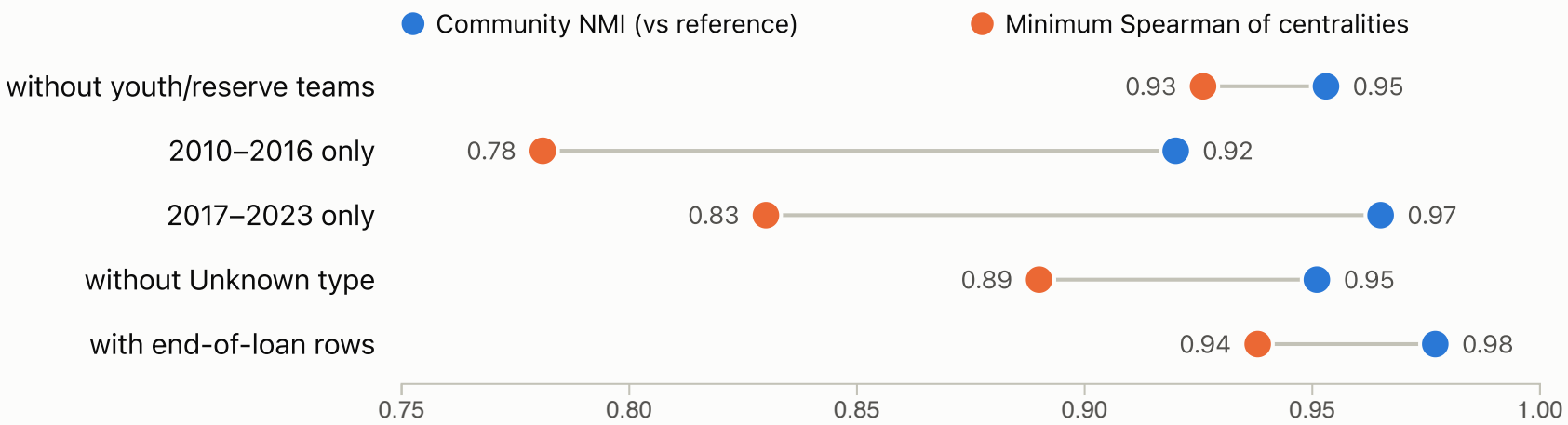
15 academies: first club is a bridge (P ≥ 0.6, 256 departures) → **5.5%** reach the elite within 3 years, against **1.2%** otherwise (1,598)
association, not causation

Capture ratio = Σ fees received at departure / Σ highest fees paid later; log scale; 95% bootstrap CI.

Source: results/tables/talent_flow_academies.csv, results/talent_flow.md

Do our results hold?

Five variants of the model compared with the reference graph



What moves: the exact top ranks → talk about club **profiles**, not a podium.

Transfermarkt coverage | Zero or hidden fees | 2010–2023 window | Transfers aggregated into a weight

Source: results/tables/sensitivity_stability.csv, results/sensitivity.md

So, who is central, and who trades with whom?

Several roles, not one centre: volume (Italy, Brazil), accessibility (Balkans), bridges between blocs (Benfica, Salzburg)

The country first, then regional blocs

Observed	Our reading	Association
direction of flows	language and history	passing through a bridge club and reaching the elite

No causal effect established.

Thank you — questions?

github.com/aarensberg/football-transfer-network-analysis